IOTAFLOW
SYSTEMS PVT. LTD

Microstream Flowsensor

OF-Z

Viscous Fluid is Measurable

Ideal to dispense Ad-blue

Excellent Repeatability

Optimal Control for Fuel such as Kerosene and heavy Oil

RoHS CE

Microstream flowsensor OF-Z is ideal for measuring a variety of liquids and it is capable of measuring small flow accurately.

Aichi tokei

Specification

Model			OF05ZAT-AR/MR	OF10ZAT-AR/MR	OF05ZZT-AR/MR	OF10ZZT-AR/MR
Nominal diameter			5mm	10mm	5mm	10mm
Accuracy guaranteed flow range	Liquid viscosity 0.3~0.8mPa·s		0.085~0.85L/min	0.7~5L/min	0.085~0.85L/min	0.7~5L/min
	Liquid viscosity 0.8~2.0mPa·s		0.05~0.85L/min	0.35~5L/min	0.05~0.85L/min	0.35~5L/min
	Liquid viscosity 2.0~5.0mPa·s		0.017~0.85L/min	0.17~5L/min	0.017~0.85L/min	0.17~5L/min
	Liquid viscosity 5.0~200mPa·s		0.0085~0.85L/min	0.085~5L/min	0.0085~0.85L/min	0.085~5L/min
Accuracy			±2%RS (Standard installation position)			
Measurable fluid ※1			Hot and cold water/Kerosene/Light oil/heavy oil		Weak acid/weak alkaline/hot and cold water/ kerosene/light oil/heavy oil	
Maximum working pressure			0.5MPa (Liquid temperature at 20℃)			
Pressure loss(@ accuracy guaranteed maximum flow)			4 kPa or less	10 kPa or less	4 kPa or less	10 kPa or less
Liquid viscosity range			0.3~200mPa·s			
Liquid temperature range			-10~+70℃ (No freezing)			
Ambient working temperature / humidity range			-10~+70℃ 35~85%RH (No condensation)			
Output signal	Voltage pulse output (Z□T-AR)	Pulse specification	Load resistance 10kΩ or more Duty ratio 2 : 8 < ON : OFF < 8 : 2			
		Cable	Lead cable length: Approx 480mm 3-wire AWG26 Flat cable			
	NPN open collector pulse output (Z□T-MR)	Pulse specification	Maximum load: 24V DC 6mA DC Duty ratio 2 : 8 < ON : OFF < 8 : 2			
		Cable	Lead cable length: Approx 600mm 4-wire AWG26 Flat cable			
Pulse unit			0.46mL/P	2.5mL/P	0.46mL/P	2.5mL/P
Maximum frequency(@ accuracy guaranteed maximum flow)			Approx 31Hz	Approx 34Hz	Approx 31Hz	Approx 34Hz
Minimum pulse ON time			Approx 6.5 ms	Approx 6 ms	Approx 6.5 ms	Approx 6 ms
Power supply			3~24VDC In case of open collector pulse output, applied voltage of sensor power supply (Red-Black) and of pulse output (Blue·White-Black) shall be the same. ※2			
Consumption current			8.4mA or less			
Installation position			Position that the nameplate is vertical to the floor (Air intrusion shall be avoided.)			
Protection structure			Indoor use (IPX4 equivalent)			
Connection			R1/4	R1/2	R1/4	R1/2
Weight			Approx 100g	Approx 140g	Approx 100g	Approx 140g
Main materials of wetted part	Case		PPS			
	Rotor		PPS			
	O-ring		NBR		FKM	
	Shaft		SUS304		SiC	

Applied voltage	High	Low
3V DC	2V DC or more	1V DC or less
12V DC	10V DC or more	
24V DC	20V DC or more	

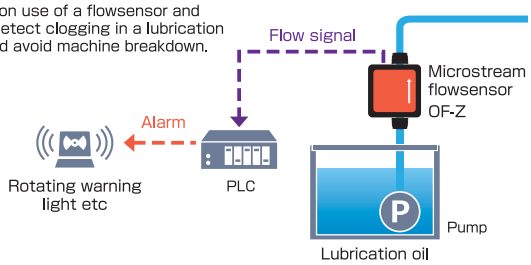
- For the details of material marks, refer to the back cover.
- If a fluid can get mixed with particles, install a filter of which mesh is #80 or more at an upstream side of flowsensor.
- Measurement of gasoline, sodium hydroxide (Caustic soda), hydrogen peroxide solution (Oxydol) and hydrochloric (strong acid) is not allowed.
- For the viscosity unit, refer to the back cover.
- Make sure to confirm product specification before usage.

※1: For weak acid and alkaline fluids, please confirm corrosion resistance of main materials of wetted part.
※2: In case applied voltage of sensor power supply and pulse output is not the same, performance and output can be unstable.

Application examples

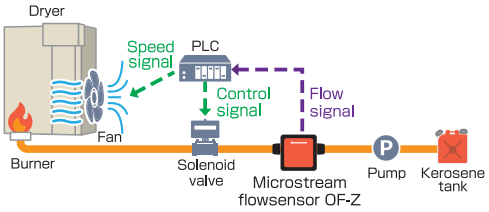
1. For flow check of lubrication oil

Combination use of a flowsensor and PLC can detect clogging in a lubrication oil pipe and avoid machine breakdown.



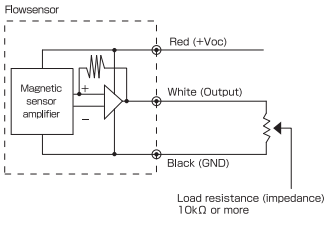
2. For monitoring of dryer's kerosene

Combination use of a flowsensor, a pump, a solenoid valve, PLC can control dried condition inside a dryer by controlling fan speed in accordance with kerosene supply amount.



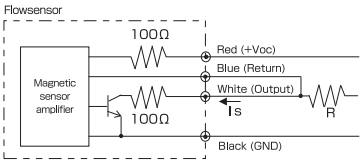
Wiring instruction

【Voltage pulse output】(Z□T-AR)



Connect under load resistance (impedance) between White-Black as 10kΩ or more

【NPN open collector pulse output】(Z□T-MR)

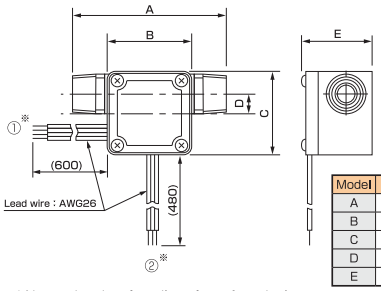


Pull-up resistor on the open collector output side showld be 50Ω or less. However, please arrange so that output sink current should be less than 6mA.

$$\text{Is (Output sink current : mA)} = \frac{\text{V (Power supply voltage: Volt)}}{\text{Current limiting resistor}} \leqq 6\text{mA}$$

Applied voltage of sensor power supply (Red-Black) and pulse output (Blue·White-Black) shall be the same.

External dimensions



Unit: mm

Model	OF05	OF10
A	80	90
B	46.9	46.9
C	46.9	46.9
D	8	8.5
E	27.3	40.3

*About the drawing direction of lead wire
(1) for open collector output type and (2) for voltage pulse output type

Manufactured and Distributed by

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Technical specifications in this catalog are up-to-date as of June 2020.

To Our Customers

Please understand that product specifications may be changed without notice in order to improve performance. We are always happy to provide the latest catalogs and brochures, and respond to inquiries made to our offices

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